

Education

University of Maryland, College Park

Ph.D., Computer Science

Advisor: *Prof. Furong Huang*

Starting Aug. 2026

College Park, MD

Daegu Gyeongbuk Institute of Science and Technology (DGIST)

B.E., College of Transdisciplinary Studies

Feb. 2019 – Feb. 2023

Daegu, South Korea

Publications (*: Equal Contribution)

Conference Proceedings

1. **Tortoise and Hare Guidance: Accelerating Diffusion Model Inference with Multirate Integration**
Yunghee Lee, Byeonghyun Pak, Junwha Hong, Hoseong Kim
Conference on Neural Information Processing Systems (NeurIPS), 2025. [\[paper\]](#)
2. **Textual Query-Driven Mask Transformer for Domain Generalized Segmentation**
Byeonghyun Pak, Byeongju Woo*, Sunghwan Kim*, Dae-hwan Kim, Hoseong Kim*
European Conference on Computer Vision (ECCV), 2024. [\[paper\]](#)
3. **B-spline Texture Coefficients Estimator for Screen Content Image Super-Resolution**
Byeonghyun Pak, Jaewon Lee*, Kyong Hwan Jin*
Conference on Computer Vision and Pattern Recognition (CVPR), 2023. [\[paper\]](#)
Highlight Paper (top 2.5%)

Workshop Papers

1. **4D Latent Mapping for Mobile Manipulation Policy Learning**
Sunghwan Kim, Byeonghyun Pak*, Kehan Long, Yulun Tian, Nikolay Atanasov*
ICRA Workshop on Semantics for Reliable Robot Autonomy, 2026.
ICRA Workshop on Multi-Modal Spatial AI, 2026. [\[paper\]](#)
Oral Presentation & Best Paper Award
2. **Rethinking FID Through the Geometry of the Reference Dataset**
Yunghee Lee, Byeonghyun Pak
ICML Workshop on Combining Theory and Benchmarks, 2026. [\[paper\]](#)
3. **Pixel-level Scene Understanding in One Token: Visual States Need What-is-Where Composition**
Seokmin Lee, Yunghee Lee, Byeonghyun Pak, Byeongju Woo
CVPR Workshop on Pixel-level Video Understanding in the Wild, 2026. [\[paper\]](#)

Preprints

1. **SERF: Spatiotemporal Environment and Robot Feature Map for Long-Horizon Mobile Manipulation**
Sunghwan Kim, Byeonghyun Pak*, Kehan Long, Yulun Tian, Nikolay Atanasov*
Under Review (2026). [\[paper\]](#)
2. **Aligning Forest and Trees in Images & Long Captions for Visually Grounded Understanding**
Byeongju Woo, Zilin Wang, Byeonghyun Pak, Sangwoo Mo, Stella X. Yu
Under Review (2026). [\[paper\]](#)

Research Experience

Independent Research

Remote Collaboration with UC San Diego

Advisor: *Prof. Nikolay Atanasov*

Jun. 2025 – Jun. 2026

Daejeon, South Korea

- Led research on Vision-Language-Action (VLA) models for long-horizon mobile manipulation.
- Proposed 4D latent mapping as a persistent spatiotemporal memory representation for policy learning.
- Presented the work at an ICRA 2026 workshop, where it received the *Best Paper Award*. [\[project page\]](#)

Agency for Defense Development
Research Officer for National Defense (Military Service)
Advisor: *Dr. Hoseong Kim*

Jun. 2023 – May 2026
Daejeon, South Korea

- Led research on vision-language models for domain-generalized segmentation. [[project page](#)]
- Studied efficient diffusion models for synthetic dataset generation. [[project page](#)]
- Contributed to visual representation learning for robotics. [[project page](#)]

Image Processing Lab, DGIST
Undergraduate Research Intern
Advisor: *Prof. Kyong Hwan Jin*

Dec. 2021 – Feb. 2023
Daegu, South Korea

- Led research on implicit neural representations for image super-resolution.
- Published the work at CVPR 2023, where it was selected as a *Highlight Paper*. [[project page](#)]

Honors & Awards

University of Maryland Dean's Fellowship	2026
• Awarded to outstanding incoming Ph.D. students	
Best Paper Award, ICRA Workshop on Multi-Modal Spatial AI	2026
• Awarded for outstanding research presented at the workshop	
Korea National Scholarship of Excellence in Science and Technology	2021 – 2023
• National full-tuition scholarship for outstanding undergraduate students	
Korea National Scholarship for Undergraduate Study	2019 – 2023
• National full-tuition scholarship with living stipend	
Korea Military Academy Superintendent's Award	2021
• Awarded for a research project on object-tracking models for defense systems	

Academic Services

Journal Reviewer	
• IEEE Transactions on Medical Imaging (TMI)	
	2025
Conference Reviewer	
• Conference on Neural Information Processing Systems (NeurIPS)	
	2025, 2026
• Conference on Computer Vision and Pattern Recognition (CVPR)	
	2026
• Conference on Robot Learning (CoRL)	
	2026
Workshop Reviewer	
• ICRA Workshop on Semantics for Reliable Robot Autonomy	
	2026

Skills

- **Programming Languages:** Python, C/C++, JavaScript, MATLAB
- **Frameworks & Tools:** PyTorch, TensorFlow, Docker, Git, OpenCV, OpenGL, Open3D
- **Languages:** Korean (native), English (fluent)